

实行三模态 RTK 定位导航

odom+imu+RTK

计划使用双卡滤波，RTK 精准定位，ODOM 与 imu 平滑辅助的功能

总体导航采用雷达建图，航点导航

RTK-GNSS：提供全局绝对位置（厘米级精度），但不连续、有延迟，且在遮挡环境下会失锁。

IMU（惯性测量单元）：提供高频姿态和角速度、加速度信息，短期精确但长期漂移。

轮式里程计（Odom）：提供连续的速度和位移估计，高频但受轮胎打滑、地面不平影响，漂移严重。

RTK 提供“锚点”（全局绝对坐标），IMU 提供“姿态”（方向和加速度），Odom 提供“连续骨架”（平滑的轨迹形状）。三者互补，通过双 EKF 滤波框架融合，输出既全局准确又局部平滑的位姿估计。

目前我们在 EKF 中只使用里程计的线速度 v_x 和角速度 v_z ，因为轮式里程计的 yaw 角漂移严重

首先插上 RTK 的移动端，C 口要插小板子，SB 陈星宇插错了导致我识别不到 CH340

而且要把中介拿掉，不然会串，因为中介也是 CH340

ros2 launch nmea_navsat_driver um982.launch.py 启动

话题列表为

```
cat@lubancat:~$ ros2 topic list
```

```
/fix
```

```
/heading
```

```
/parameter_events
```

```
/rosout
```

```
/time_reference
```

```
/vel
```

```
cat@lubancat:~$
```

打开 imu 和 odom 和雷达

```
cd ~/ranger_ws/src/ranger_ros2/ranger_bringup/scripts
```

```
sudo bash setup_can2usb.bash
```

```
ros2 launch ranger_bringup ranger_mini_v3.launch.py
```

```
ros2 launch rslidar_sdk start.py
```

```
ros2 run rs_to_velodyne rs_to_velodyne_node
```

```
ros2 launch hipnuc_imu imu_spec_msg.launch.py
```

```
cat@lubancat:~$ ros2 topic list
```

```
/IMU_data
```

```
/actuator_state
```

```
/battery_state
```

```
/cmd_vel
```

```
/euler_data
```

```
/fix
```

```
/heading
```

```
/magnetic_data
```

```
/motion_state
```

```
/odom
```

```
/parameter_events
```

```
/rosout
```

```
/rslidar_points
```

```
/system_state
```

```
/tf
```

```
/time_reference
/vel
/velodyne_points
cat@lubancat:~$
```

RTK 原始数据

```
[nmea_serial_driver-1] [INFO] [1779254089.826826673] [um982_driver]: [RAW]:
$GNRMC,051450.00,A,4544.40570896,N,12643.03778429,E,0.055,102.6,200526,10.5,W,A,C*67
[nmea_serial_driver-1] [INFO] [1779254089.830328715] [um982_driver]: [RAW]:
$GPHDT,0.0000,T*05
[nmea_serial_driver-1] [INFO] [1779254090.818632273] [um982_driver]: [RAW]:
$GNGGA,051451.00,4544.40571296,N,12643.03784459,E,1,19,0.7,147.1690,M,14.4363,M,,*7B
[nmea_serial_driver-1] [INFO] [1779254090.827055024] [um982_driver]: [RAW]:
$GNRMC,051451.00,A,4544.40571296,N,12643.03784459,E,0.073,193.4,200526,10.5,W,A,C*67
```

启动 RTK 双卡定位

```
ros2 launch rtk_localization rtk_localization.launch.py
```

如果启动不了

就在工作空间中

```
source install/setup.bash ros2 launch rtk_localization rtk_localization.launch.py
```

启动我们自己制作的

```
ros2 topic echo /odometry/gps
```

查看原始 ros 数据

```
ros2 topic echo /fix
```

在出去测试之前，我需要知道我们各传感器的相对位置，组成一个属于小车底盘中心与传感器之间的 tf

```
cat@lubancat:~$ ros2 run tf2_ros tf2_echo base_footprint rslidar
[INFO] [1779257680.408615598] [tf2_echo]: Waiting for transform base_footprint -> rslidar: Invalid frame ID "base_footprint" passed to canTransform argument target_frame - frame does not exist
At time 0.0
- Translation: [-0.019, 0.000, 0.830]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.000, 1.000]
- Rotation: in RPY (radian) [0.000, -0.000, 0.000]
- Rotation: in RPY (degree) [0.000, -0.000, 0.000]
- Matrix:
1.000 0.000 0.000 -0.019
0.000 1.000 0.000 0.000
0.000 0.000 1.000 0.830
0.000 0.000 0.000 1.000
At time 0.0
- Translation: [-0.019, 0.000, 0.830]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.000, 1.000]
- Rotation: in RPY (radian) [0.000, -0.000, 0.000]
- Rotation: in RPY (degree) [0.000, -0.000, 0.000]
- Matrix:
1.000 0.000 0.000 -0.019
0.000 1.000 0.000 0.000
0.000 0.000 1.000 0.830
0.000 0.000 0.000 1.000
^C[INFO] [1779257682.975918356] [rclcpp]: signal_handler(signum=2)
cat@lubancat:~$ ros2 run tf2_ros tf2_echo base_footprint imu_link
[INFO] [1779257690.852663314] [tf2_echo]: Waiting for transform base_footprint -> imu_link: Invalid frame ID "base_footprint" passed to canTransform argument target_frame - frame does not exist
At time 0.0
- Translation: [-0.087, 0.000, 0.596]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.000, 1.000]
- Rotation: in RPY (radian) [0.000, -0.000, 0.000]
- Rotation: in RPY (degree) [0.000, -0.000, 0.000]
- Matrix:
1.000 0.000 0.000 -0.087
0.000 1.000 0.000 0.000
0.000 0.000 1.000 0.596
0.000 0.000 0.000 1.000
At time 0.0
- Translation: [-0.087, 0.000, 0.596]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.000, 1.000]
- Rotation: in RPY (radian) [0.000, -0.000, 0.000]
- Rotation: in RPY (degree) [0.000, -0.000, 0.000]
- Matrix:
```

```
1.000 0.000 0.000 -0.087
0.000 1.000 0.000 0.000
0.000 0.000 1.000 0.596
0.000 0.000 0.000 1.000
^C[INFO] [1779257693.357793816] [rclcpp]: signal_handler(signum=2)
cat@lubancat:~$
```

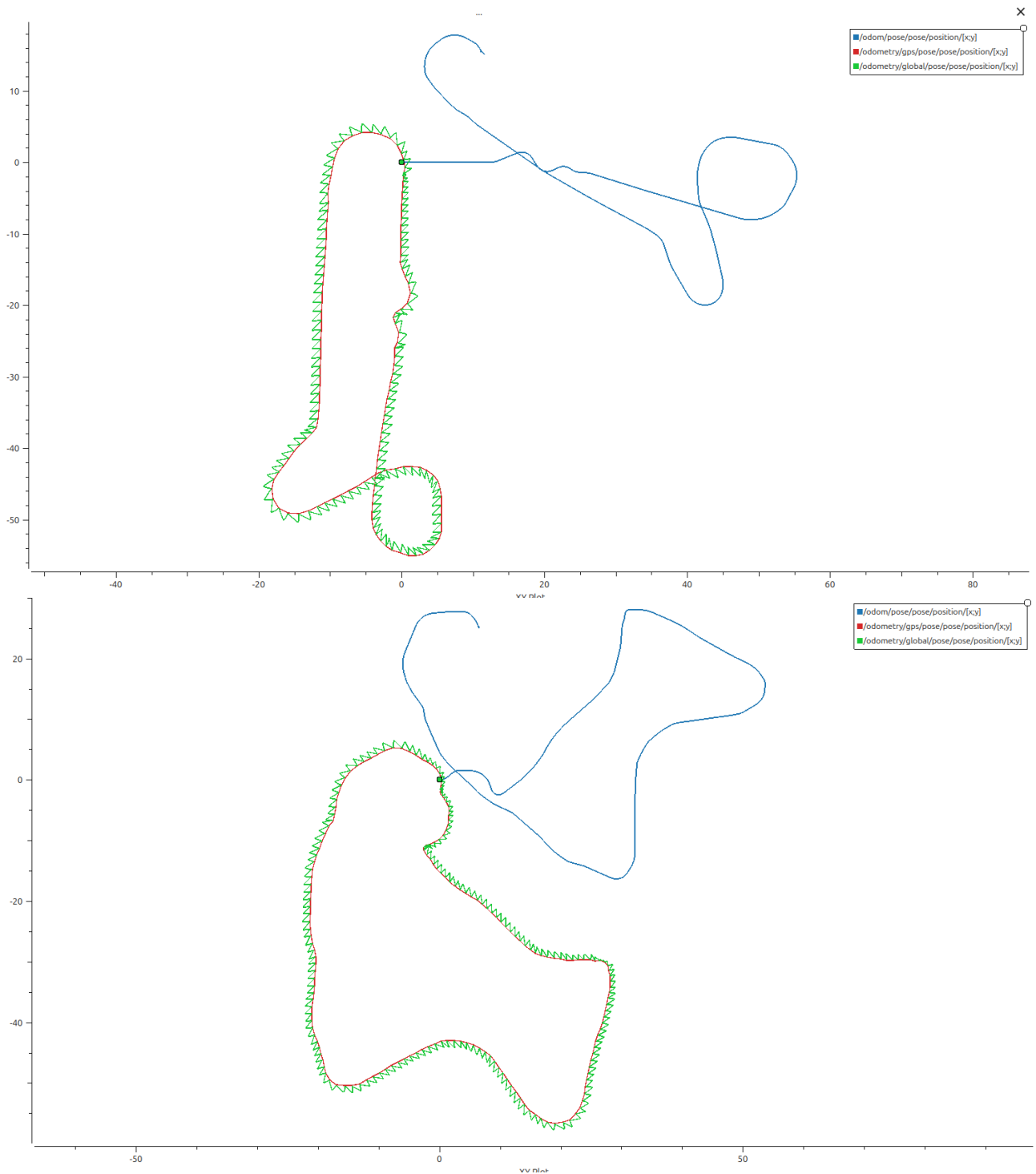
```
ros2 bag record -o rtk_test_0520 \
/fix \
/heading \
/odom \
/IMU_data \
/rslidar_points \
/tf \
/tf_static \
/odometry/gps \
/odometry/local \
/odometry/global
```

录制

```
cd /home/cat/rtk_verify_ws/bags
ros2 bag record -o trajectory_comparison1 /odom /odometry/gps /odometry/global /tf /tf_static
```

使用这个工具查看轨迹

```
ros2 run plotjuggler plotjuggler
```



根据这个图像，我得知，

[纬度, 经度, 初始航向角] -> -1.5707963

代表开机车头朝向正南 datum: [45.7394912333, 126.7170297150, -1.5707963]

因此我改进

此时为了验证改进后的效果

1.再跑一圈，看轨迹

2.静止时抓 imu 的零点朝向

ros2 topic echo /IMU_data -once

[illegible]

[illegible]

- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0

开回原点大致位置

ros2 topic echo /odometry/gps --once

header:

stamp:

sec: 1779292765

nanosec: 437755599

frame_id: map

child_frame_id: "

pose:

pose:

position:

x: -0.0319956829778838

y: 0.2921979754736149

z: 0.0

orientation:

x: 0.0

y: 0.0

z: 0.0

w: 1.0

covariance:

- 0.0001

- 9.390921076858088e-23

- 0.0

- 0.0

- 0.0

- 0.0

- 1.8947645956639246e-22

- 0.0001

- 0.0

- 0.0

- 0.0

- 0.0

- 0.0

- 0.0

- 0.0

- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0

3.我想要在小车启动后让小车朝南方走几米，抓取 odom 与 imu 与 gps 与 heading 的三个输出数据，然后以 txt 形式存在/home/cat/rtk_verify_ws/bags 里

cd /home/cat/rtk_verify_ws/bags

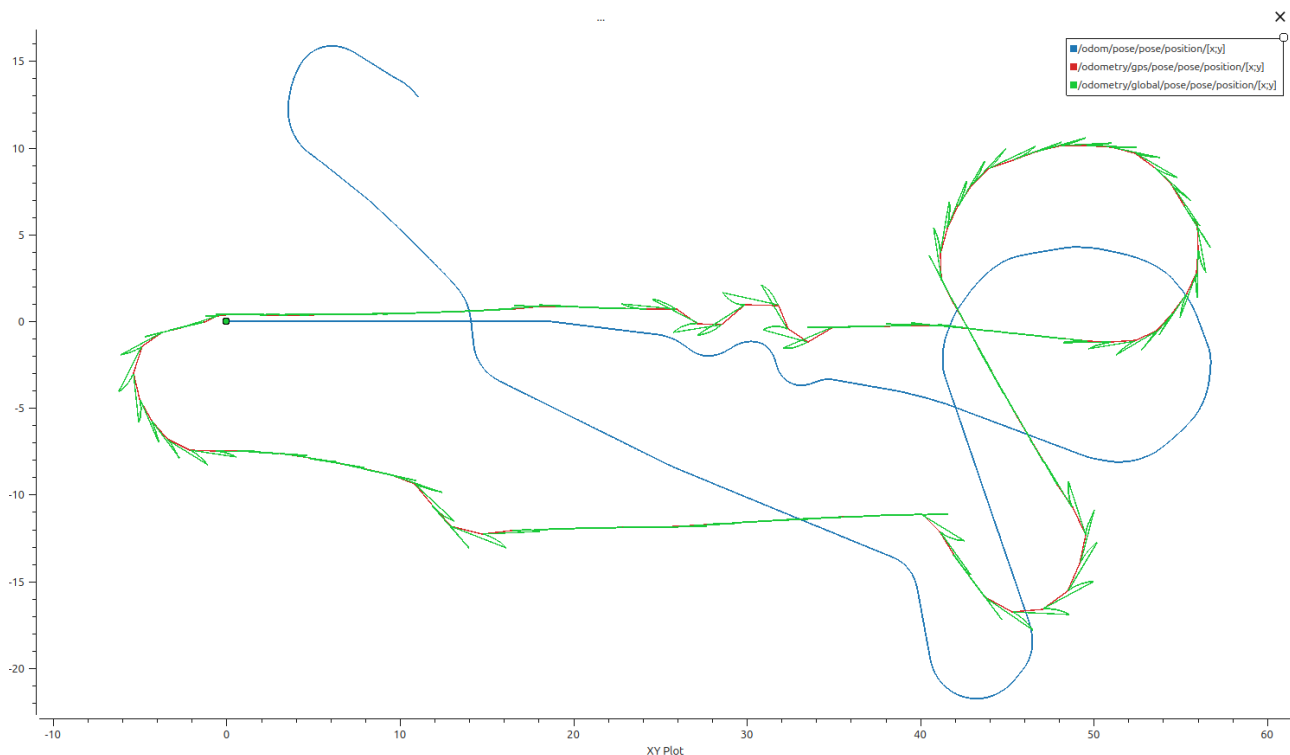
然后

ros2 topic echo /odom > odom.txt

ros2 topic echo /IMU_data > imu.txt

ros2 topic echo /heading > heading.txt

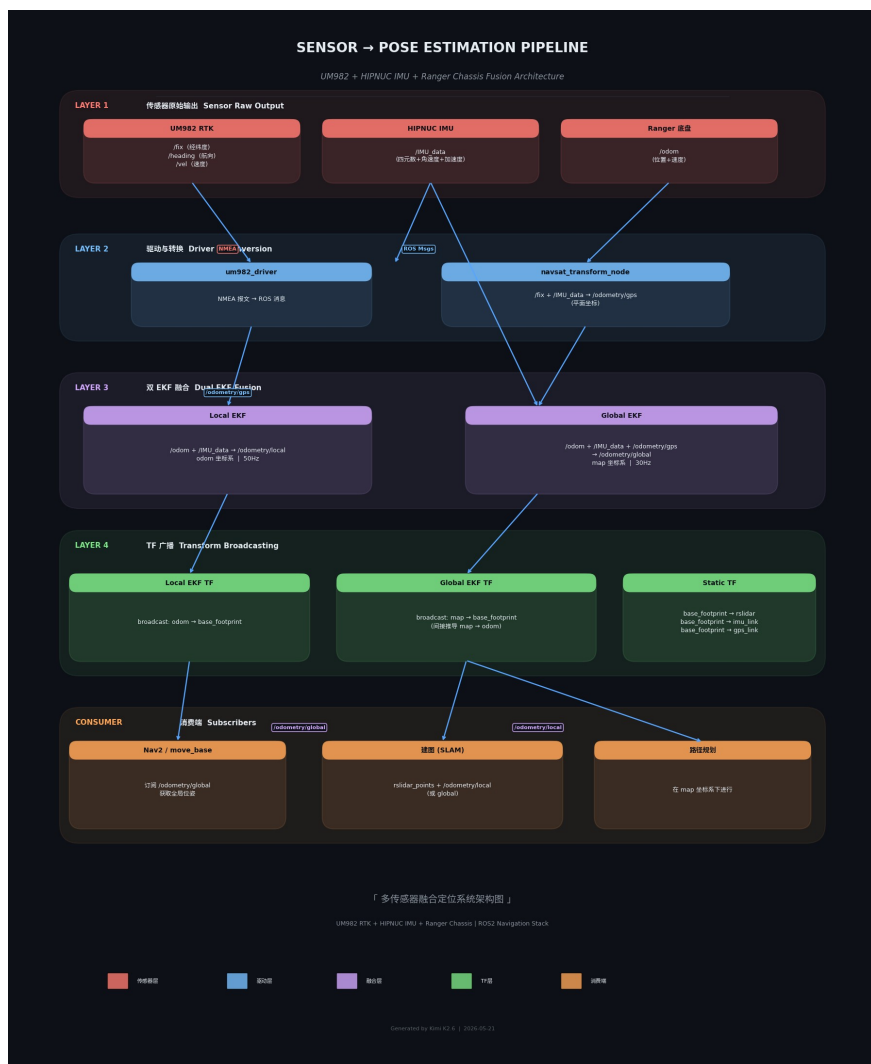
ros2 topic echo /fix > gps.txt



方向问题已解决

发现两个问题

1. 轮式里程计漂移严重
2. 为什么双卡滤波后的绿色轨迹不如纯 gps 的蓝色轨迹平滑



从原始传感器数据到最终位姿估计：

【第一层：传感器原始输出】

- UM982 RTK → /fix (经纬度) + /heading (航向) + /vel (速度)
- HIPNUC IMU → /IMU_data (四元数+角速度+加速度)
- Ranger 底盘 → /odom (位置+速度)

【第二层：驱动与转换】

- um982_driver: NMEA 报文 → ROS 消息
- navsat_transform_node: /fix + /IMU_data → /odometry/gps (平面坐标)

【第三层：双 EKF 融合】

- Local EKF: /odom + /IMU_data → /odometry/local (odom 坐标系, 50Hz)
- Global EKF: /odom + /IMU_data + /odometry/gps → /odometry/global (map 坐标系, 30Hz)

【第四层：TF 广播】

- Local EKF 广播: odom → base_footprint
- Global EKF 广播: map → base_footprint (间接推导 map → odom)
- static_transform_publisher 广播: base_footprint → rslidar / imu_link / gps_link

【消费端】

- Nav2 / move_base: 订阅 /odometry/global 获取全局位姿
- 建图 (SLAM): 使用 rslidar_points + /odometry/local (或 global)
- 路径规划: 在 map 坐标系下进行

建图步骤

```
cd ~/ranger_ws/src/ranger_ros2/ranger_bringup/scripts
```

```
sudo bash setup_can2usb.bash
```

```
ros2 launch ranger_bringup ranger_mini_v3.launch.py
```

```
ros2 launch hipnuc_imu imu_spec_msg.launch.py
```

```
ros2 launch rslidar_sdk start.py
```

```
ros2 run rs_to_velodyne rs_to_velodyne_node
```

```
ros2 launch rtk_localization mapping_absolute.launch.py
```

然后要启动 slam_toolbox

```
ros2 service call /slam_toolbox/change_state lifecycle_msgs/srv/ChangeState "{transition: {id: 3}}"
```

然后就可以建图了

启动后检查

`ros2 topic echo /fix -once`（这个命令其实没用，会报错，直接 `ros2 topic echo /fix` 倒是能看）

保存

`ros2 run nav2_map_server map_saver_cli -f ~/rtk_verify_ws/maps/absolute_parking`